

CS & IT ENGINEERING

DISCRETE MATHS
COMBINATORICS



Lecture No. 1



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TOPICS TO BE COVERED

01 Sum Rule

02 Product Rule

03 Sum Rule & Product Rule

COMBINATORICS



Sum Rule/product Rule ✓

P/C $\rightarrow$ Reptⁿ.

pigeon hole principle.

Inclusion-Exclusion.

Euler- ϕ -function ✓

* Generating function. (2024)

Recurrence Relation.

Derangement.

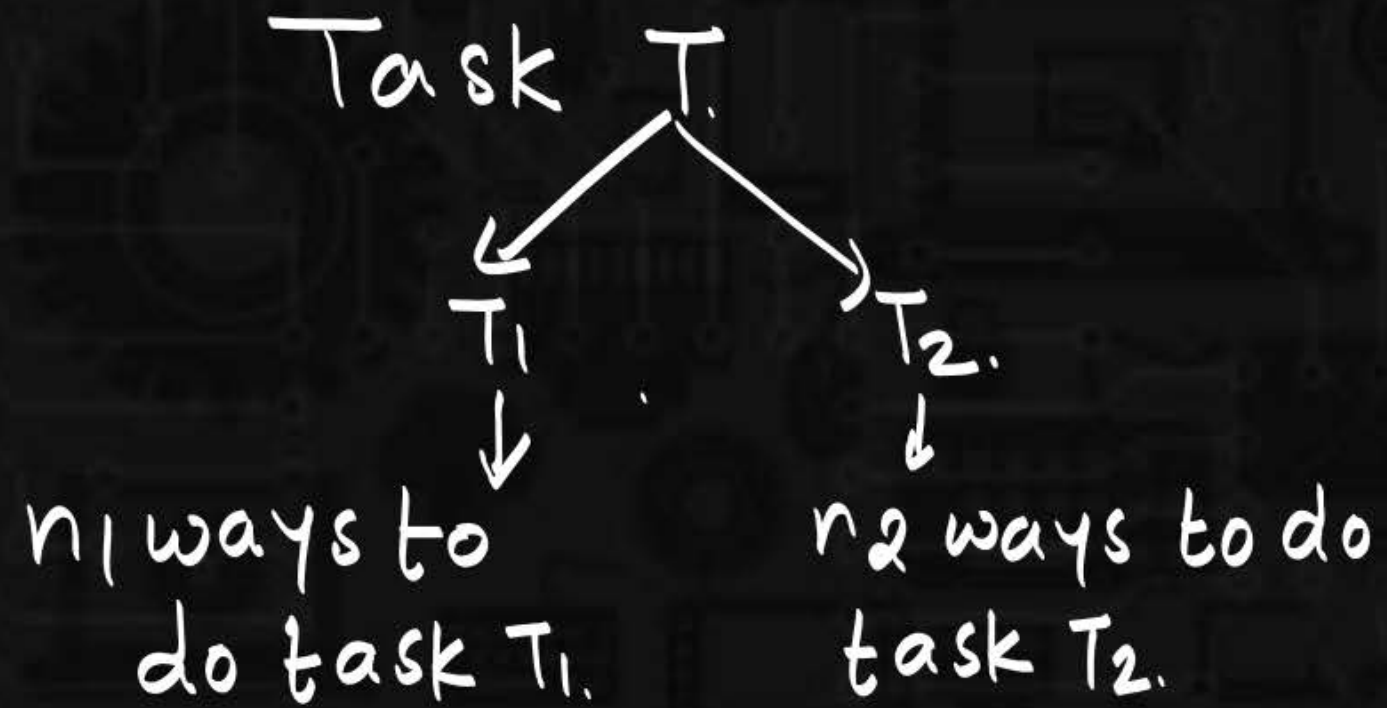
Binomial coefficient

Extended binomial coefficient

COMBINATORICS



Product Rule : (x)



* both tasks are happening simultaneously.

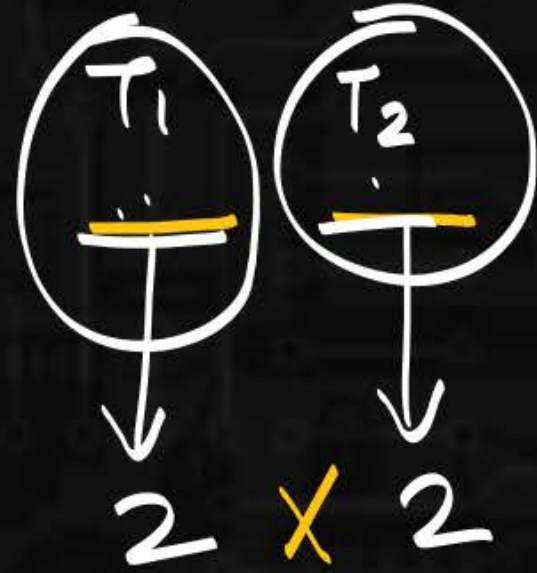
* for each of n_1 ways we can perform n_2 .

Total ways = $n_1 \times n_2$.

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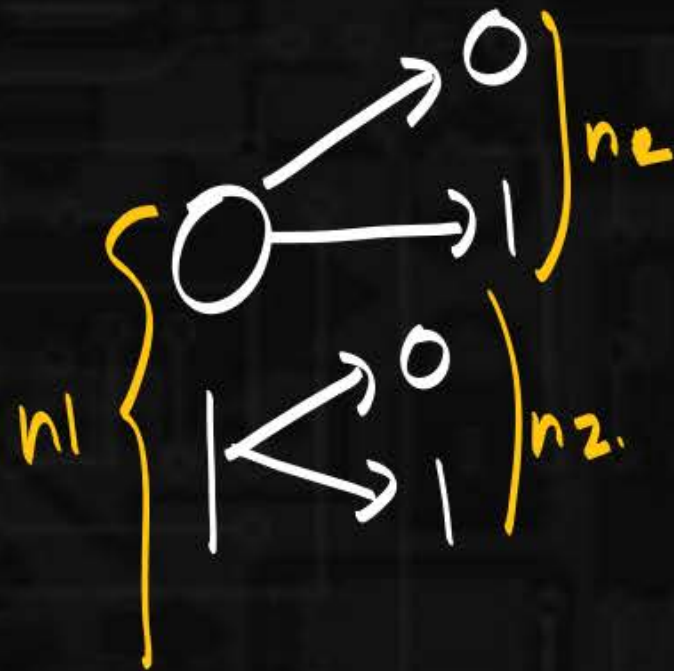
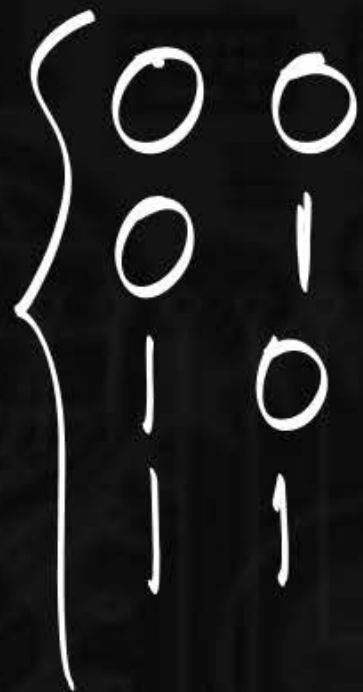


T: how many ways we can generate **signal** if we have 2 bit **code** ?



2 ways $\rightarrow T_1 \begin{cases} 0 \\ 1 \end{cases}$
2 ways $\rightarrow T_2 \begin{cases} 0 \\ 1 \end{cases}$

$$\text{Total ways} = 2 \times 2 = 2^2 = 4.$$



COMBINATORICS

$$T \begin{cases} \begin{matrix} k=0 \\ T_1: \text{for } i^0 = 1 \longrightarrow n_1 \\ T_2: \text{for } j^0 = 1 \longrightarrow n_2. \\ k = k+1. \end{matrix} \end{cases}$$

$$\underline{K = n_1 \times n_2}$$

$$\underline{i^0 = 1} \quad j^0 = 1 \longrightarrow n_2.$$

$$\underline{i^0 = 2} \quad j^0 = 1 \longrightarrow n_2$$

$$i^0 = 3 \quad j^0 = 1 \longrightarrow n_2.$$

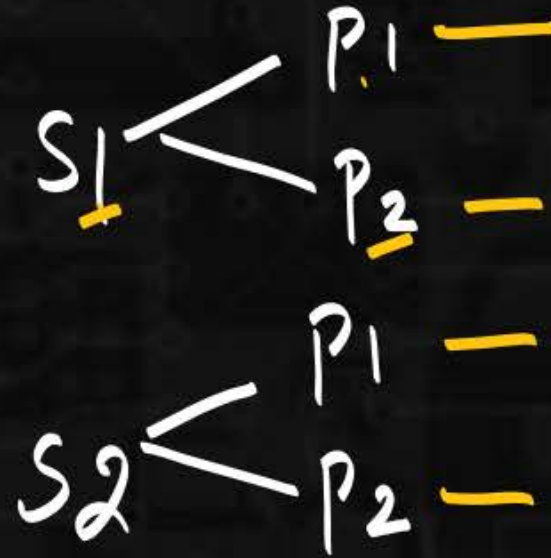
$$\vdots$$

$$i^0 = n_1$$

COMBINATORICS

how many ways i can dress up
if i have 2 shirts & 2 pants?

$\frac{2}{S_1}$	$\frac{2}{P_1}$
S_2	P_2

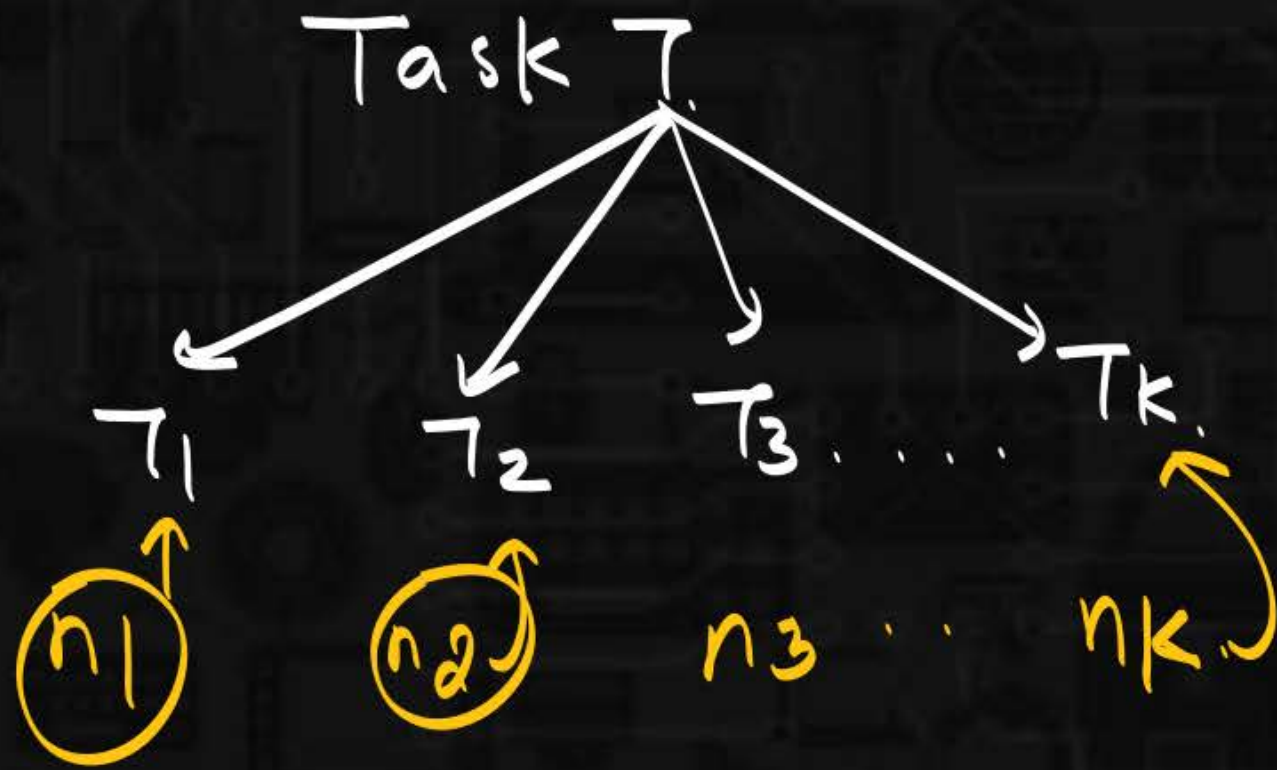


$$\text{Total ways} = 2 \cdot 2 = 4$$

COMBINATORICS



Extended product Rule:



all tasks are happening
Simultaneously.

$$\begin{aligned} \text{Total ways} \\ = n_1 \times n_2 \times n_3 \dots n_k \end{aligned}$$

COMBINATORICS



$$k = 0$$

$$K = n_1 \times n_2 \times \dots \times n_k.$$

for $i_1 = 1 \rightarrow n_1$

for $i_2 = 1 \rightarrow n_2$

for $i_3 = 1 \rightarrow n_3$

for $i_k = 1 \rightarrow n_k$

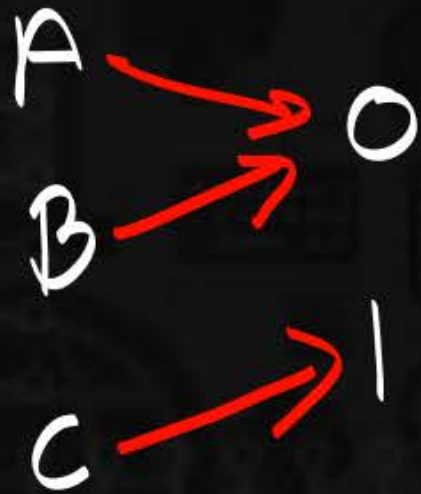
$k = k + 1$

COMBINATORICS

$$f: A \rightarrow B$$

$$|A| = 3 \quad |B| = 2$$

$$\text{Total no. of functions} = (r.s)^{L.s} = 2^3$$

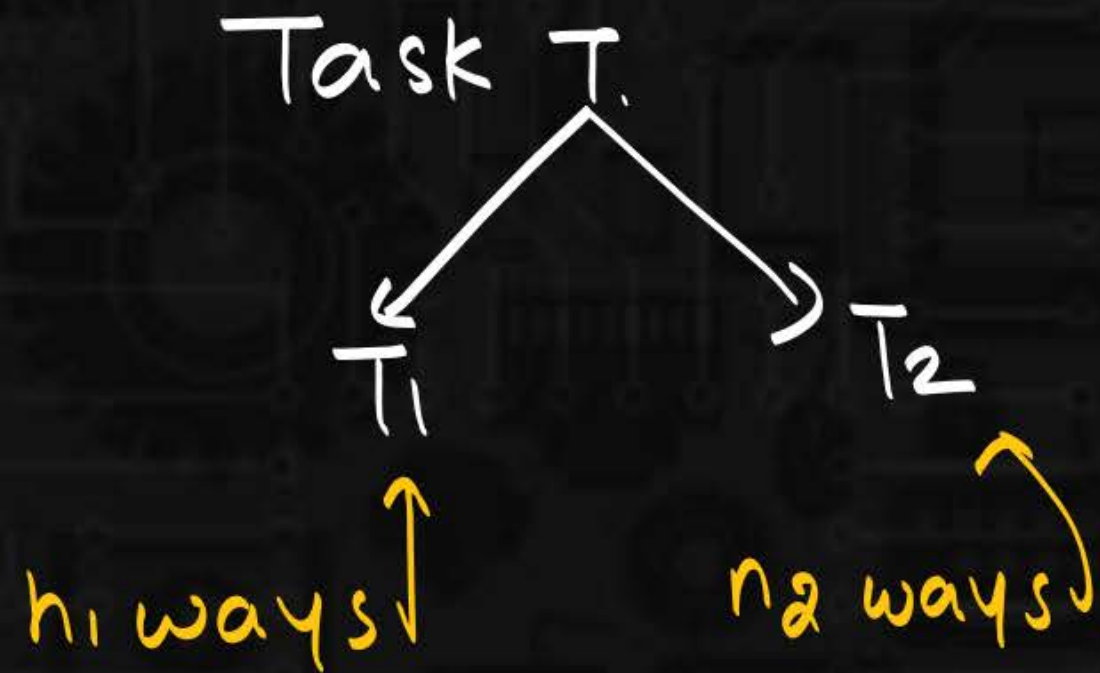


$$\frac{A}{2} \cdot \frac{B}{2} \cdot \frac{C}{2} = 2^3$$

COMBINATORICS



Sum Rule:



* both the tasks are not happening simultaneously.

$$\text{Total ways} = n_1 + n_2.$$

COMBINATORICS



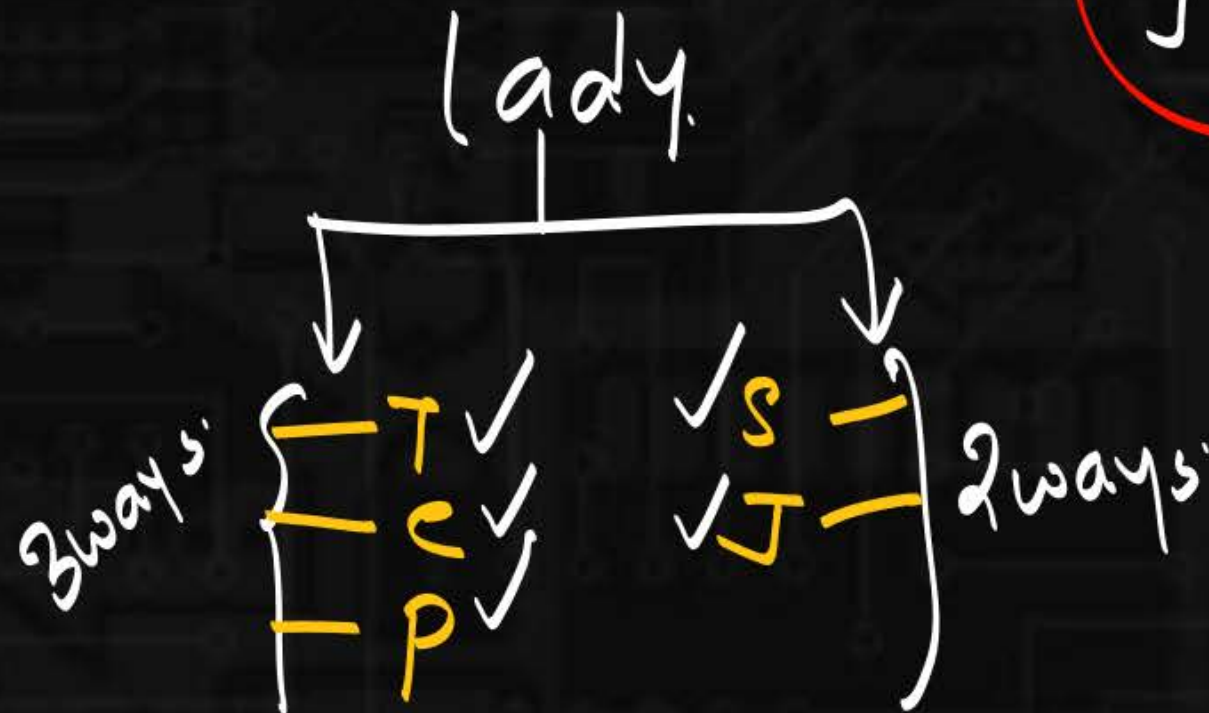
Babita ji is going for shopping.
if she does the shopping in Delhi, she can buy
She does shopping in Mumbai she buys

3 ways.

Table
Chair
Pen.

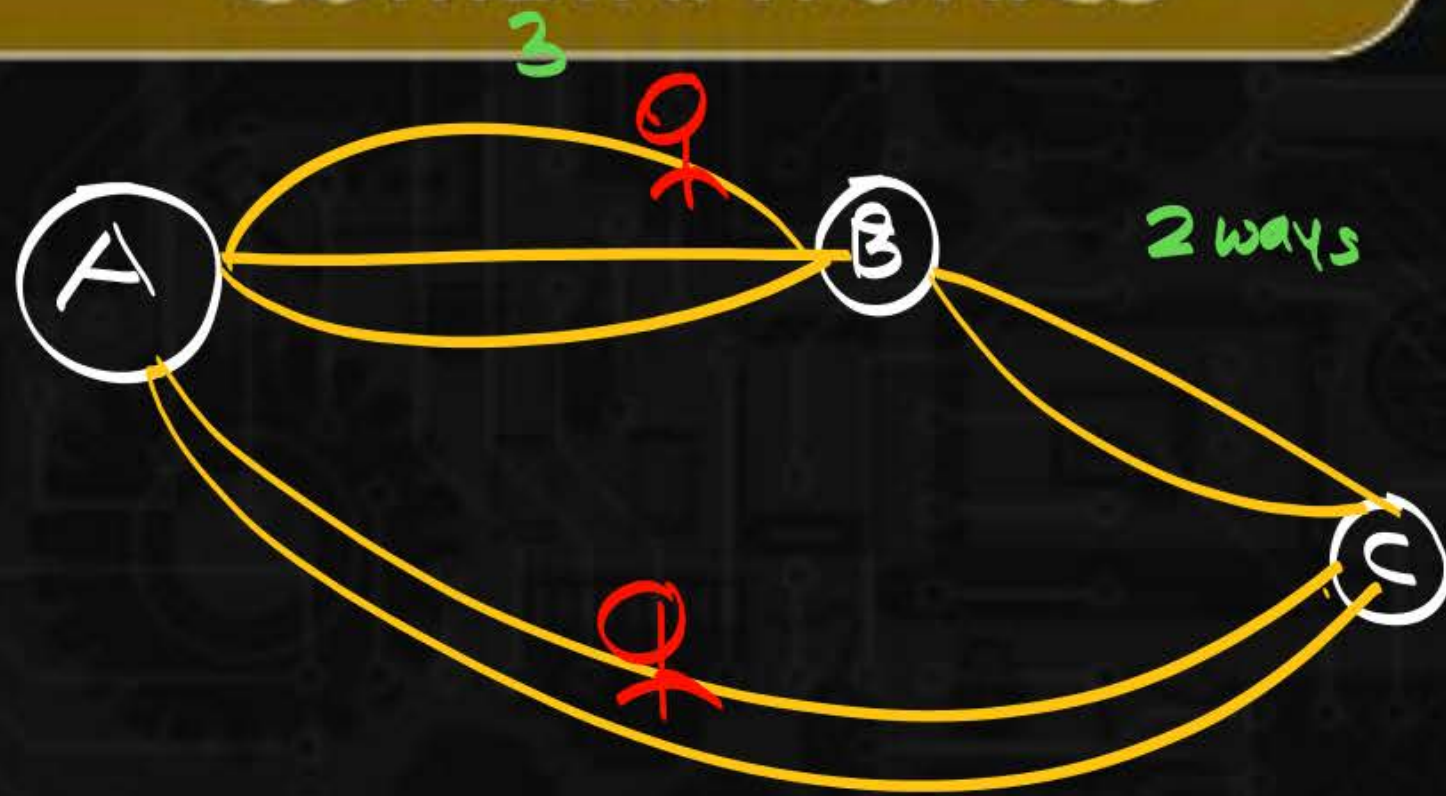
2 ways.

Saree
jeans.



$$\text{Total ways} = 3 + 2 = \underline{5 \text{ ways.}}$$

COMBINATORICS



1. $A \text{ --- } B \text{ --- } C$
 $3 \times 2 \text{ ways}$

2. $A \text{ --- } C$
 via B 3×2 + without B. 2 ways

3. $\left\{ \begin{array}{l} A \text{ --- } C \text{ --- } A \\ \text{via B} \quad \text{via B} \end{array} \right.$

4. $\left\{ A \text{ --- } C \text{ --- } A \right.$

COMBINATORICS



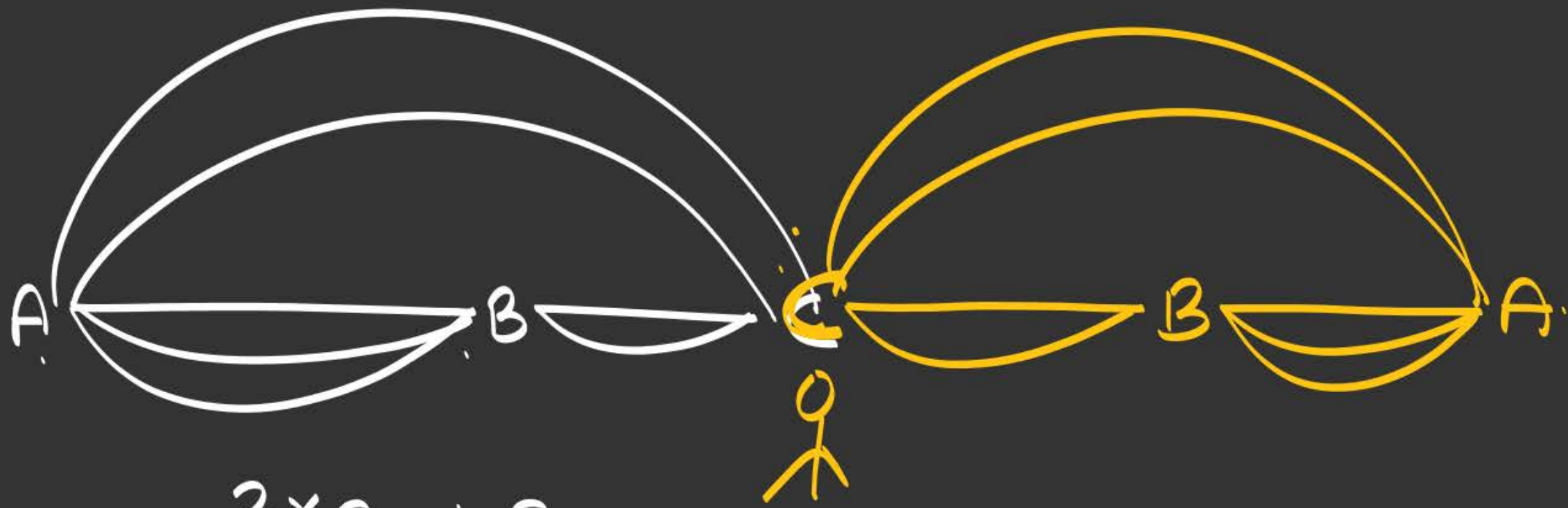
2 ways $\times$ 2 ways $\times$ 2 ways $\times$ 3 ways.

= 36 ways.

Q.4.



$8 \times 8 = 64$ ways.



$$3 \times 2 + 2$$

$$= 8 \text{ ways} \times 6 \text{ ways} + 8 \text{ ways} \times 2 \text{ ways}$$

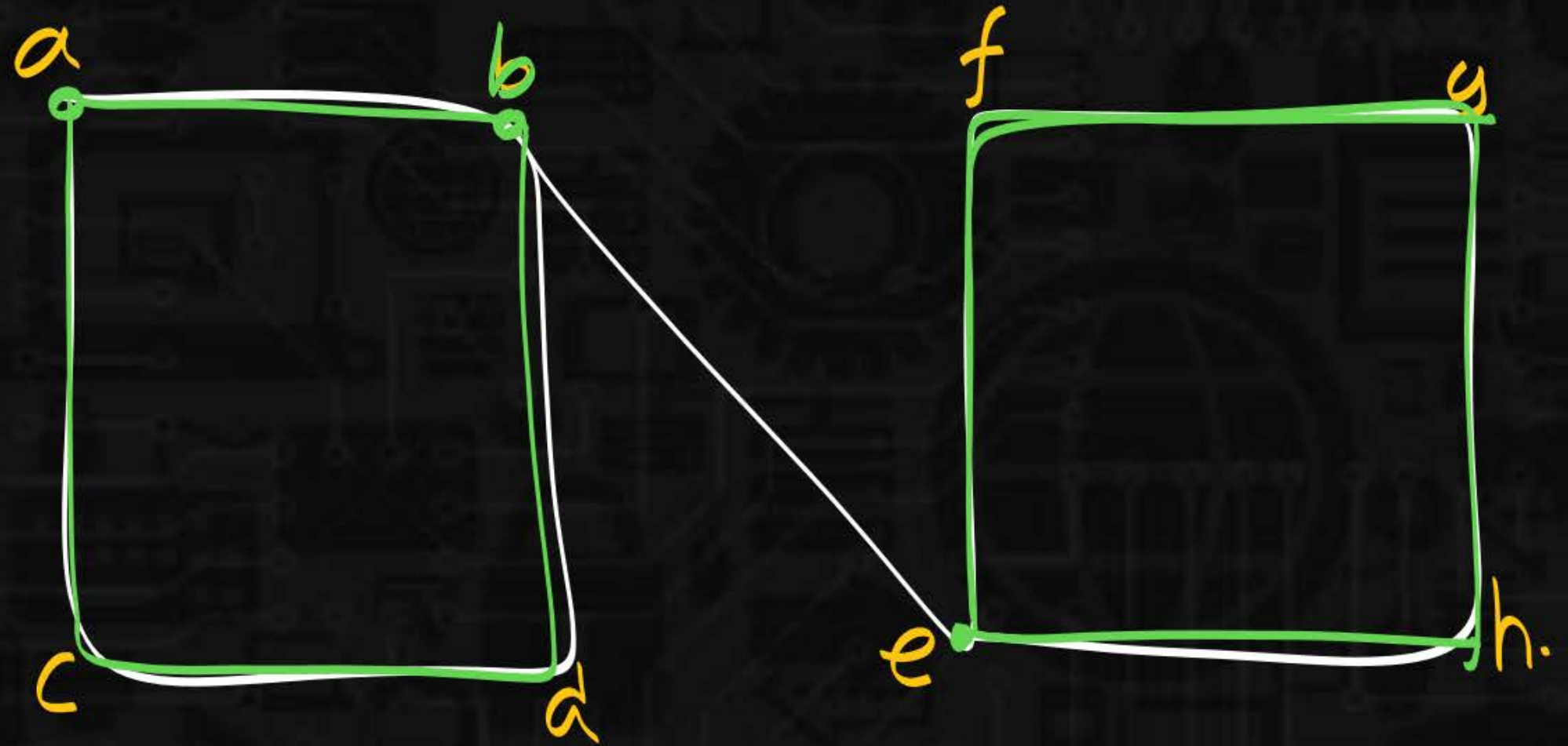
$$8 \text{ ways} (6 + 2)$$

$$= 64$$

COMBINATORICS



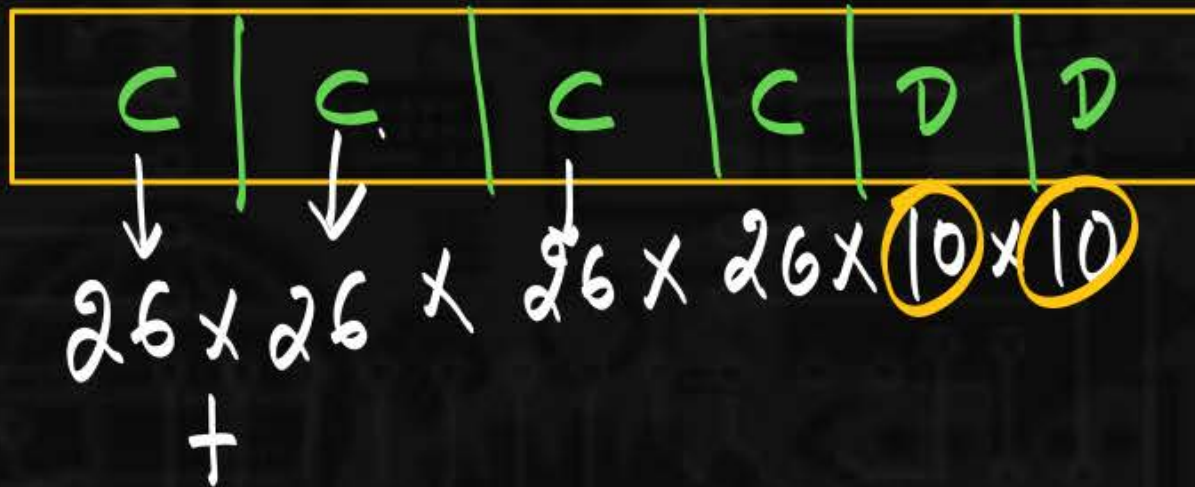
paths are
there
from a to h.



COMBINATORICS

how many number plates
we can generate if
4 characters followed by
2 digit?

$$= \underline{26^4} \times 10^2$$



COMBINATORICS



how many number plates
we can generate if
4 characters followed by
2 digit or 3 or 4 digit?

$$26^4 \cdot 10^2 + 26^4 \cdot 10^3 + 26^4 \cdot 10^4$$
$$= 26^4 (10^2 + 10^3 + 10^4)$$

C		C		C	C	D	D
---	--	---	--	---	---	---	---

$26^4 \cdot 10^2$

$\times 10$

+

C C C C D D D

$26^4 \cdot 10^3$

+

C C C C D D D D

$+ 26^4 \cdot 10^4$

COMBINATORICS



how many number plates we can
generate if 1 OR 2 OR 3 OR 4 characters followed by
1 OR 2 OR 3 OR 4 Digit?

$$(26 + 26^2 + 26^3 + 26^4)(10 + 10^2 + 10^3 + 10^4)$$

COMBINATORICS



how many 4 digit integers
are there with digit 0
appearing exactly once?

Ans: 2187

$$|S| = n$$

how many ways we can choose
subsets P and Q

$$P \cap Q = \emptyset$$

- a) 2^n
- b) $3^n - 1$
- c) $3^n + 1$
- d) 3^n

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how many integers

$\{ 1, 2, 3 \dots 1,00,000 \}$

Ans: 12600

contains exactly one 3, exactly one 4, exactly one 5?
in decimal representation?

